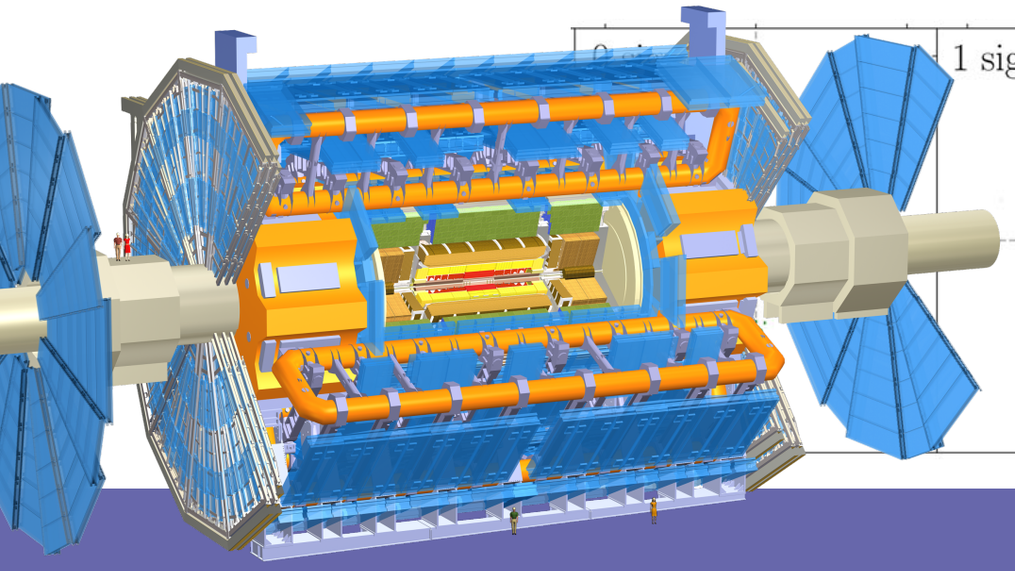
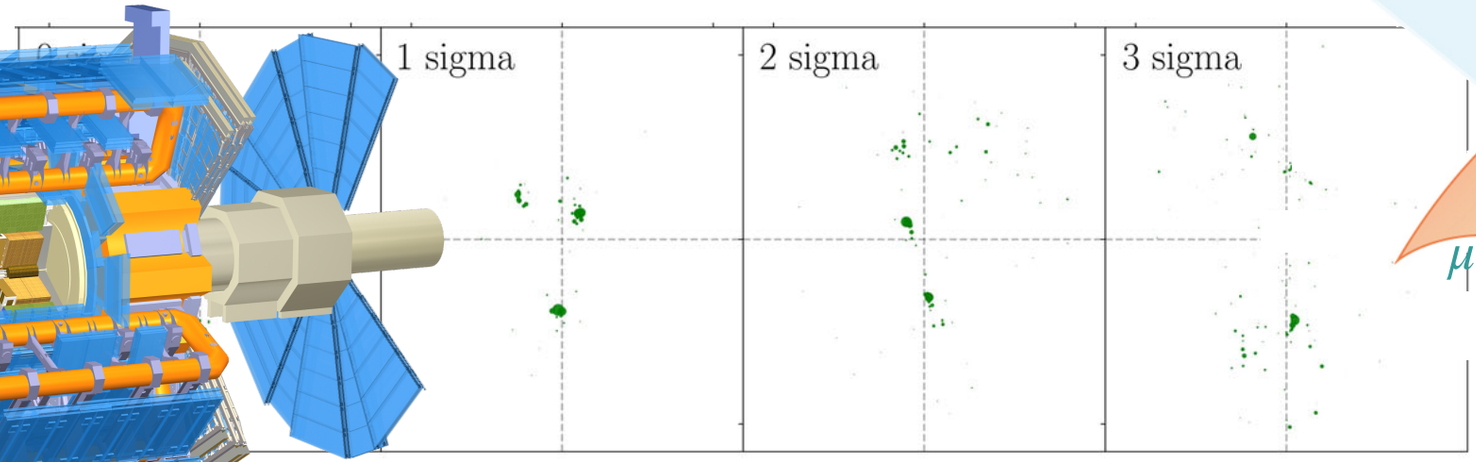
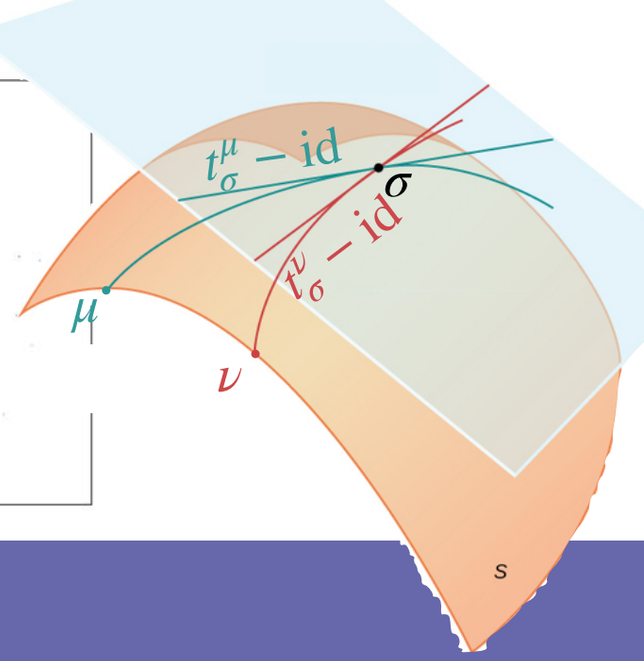






The Riemannian Structure of the 2-Wasserstein Distance: Gradient Flows and Linearization

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October 16, 2025

Plan

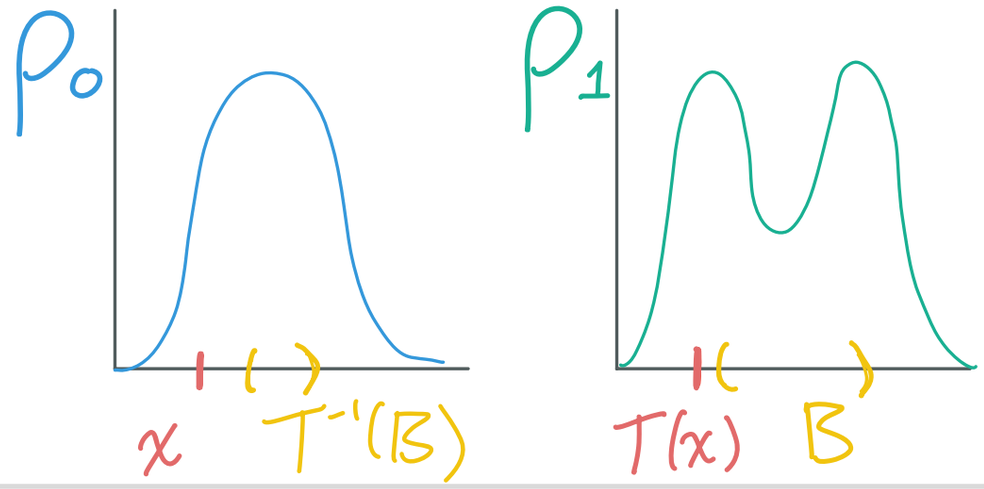
- A formal Riemannian structure for the 2-Wasserstein metric
- Application #1: Gradient flows and sampling
- Application #2: Linearized OT and jet classification

The Monge Problem

For $\rho_0, \rho_1 \in \mathscr{P}(\Omega)$, a map $\mathbf{T} : \Omega \to \Omega$ transports ρ_0 onto ρ_1 if

$$\int_\Omega f(x) d\rho_1(x) = \int_\Omega f(\mathbf{T}(x)) d\rho_0(x), \quad \forall f \in L^\infty(\Omega);$$

In this case, we write $\mathbf{T}\#\rho_0 = \rho_1$.



The 2-Wasserstein distance between $\rho_0, \rho_1 \in \mathscr{P}_{\text{ac}}(\Omega)$ is

$$W_2^2(\rho_0, \rho_1) = \inf_{\mathbf{T}} \left\{ \int_\Omega |\mathbf{T}(x) - x|^2 d\rho_0(x) : \mathbf{T}\#\rho_0 = \rho_1 \right\}$$

Let $\mathbf{T}_{\rho_0}^{\rho_1}$ denote the optimal transport map from ρ_0 to ρ_1 and $\mathbf{I}(x) = x$.

Wasserstein geodesics

Not just a metric space… a geodesic metric space:

$$\rho_t := \mathbf{T}_t \# \rho_0, \quad \mathbf{T}_t := ((1-t)\mathbf{I} + t\mathbf{T}_{\rho_0}^{\rho_1}) \ ,$$

satisfies $W_2(\rho_t, \rho_s) = |t-s| W_2(\rho_0, \rho_1)$.

Monge

Kantorovich



ρ_0 2-Wasserstein geodesic $\rho(t)$ ρ_1



ρ_0 linear interpolation $(1-t)\rho_0 + t\rho_1$ ρ_1

[Peyré, Papadakis, Oudet 2013]

Formal Riemannian structure

[Otto '01]

Goal: Riemannian structure for $M = \mathscr{P}(\Omega)$

- Tangent space $T_\rho M$ at each point $\rho \in M$
- Inner product $\langle \cdot, \cdot \rangle_\rho$ on $T_\rho M$
- Corresponding norm $\|w\|_\rho^2 = \langle w, w \rangle_\rho$ induces W_2 distance:

$$W_2^2(\rho_0, \rho_1) = \inf_{\rho_t} \left\{ \int_0^1 \|\partial_t \rho_t\|_{\rho_t}^2 dt : \rho_t|_{t=0} = \rho_0,\ \rho_t|_{t=1} = \rho_1 \right\}$$

Caveat: Formal

- Infinite dimensional
- Degenerates as we go from smooth, strictly positive measures to $\mathscr{P}_{\mathrm{ac}}(\Omega)$ and ultimately $\mathscr{P}(\Omega)$

Dynamic characterization of W_2

For any smooth curve ρ_t in $\mathscr{P}(\Omega)$, $\exists$ velocity field $\mathbf{v_t}$ so that

$$(\mathrm{CTY}) \begin{cases} \partial_t \rho_t + \nabla \cdot (\mathbf{v_t} \rho_t) = 0 \\ \rho_t|_{t=0} = \rho_0 \end{cases}$$

Ex: If $\rho_t = \mathbf{T}_t \# \rho_0$, one can show that $\mathbf{v_t} \circ \mathbf{T}_t = \mathbf{T}_{\rho_0}^{\rho_1} - \mathbf{I}$, so for all $t \in [0,1]$

$$W_2^2(\rho_0, \rho_1) = \int_\Omega |\mathbf{T}_{\rho_0}^{\rho_1} - \mathbf{I}|^2 d\rho_0 = \int_\Omega |\mathbf{v_t} \circ \mathbf{T}_t|^2 d\rho_0 = \int_\Omega |\mathbf{v_t}|^2 d\rho_t$$

Theorem: [Benamou, Brenier '00]

$$W_2^2(\rho_0, \rho_1)$$

$$= \inf_{(\rho_t, \mathbf{v_t})} \left\{ \int_0^1 \int_\Omega |\mathbf{v_t}|^2 d\rho_t \, dt : \partial_t \rho_t + \nabla \cdot (\mathbf{v_t} \rho_t) = 0, \, \rho_t|_{t=0} = \rho_0, \, \rho_t|_{t=1} = \rho_1 \right\}$$

Dynamic characterization of W_2

Theorem: [Benamou, Brenier '00]

$$W_2^2(\rho_0, \rho_1)$$

$$= \inf_{(\rho_t, \mathbf{v_t})} \left\{ \int_0^1 \int_\Omega |\mathbf{v_t}|^2 d\rho_t \, dt : \overbrace{\partial_t \rho_t + \nabla \cdot (\mathbf{v_t}\rho_t) = 0}^{\text{(CTY)}}, \; \rho_t|_{t=0} = \rho_0, \; \rho_t|_{t=1} = \rho_1 \right\}$$

Note:

- If $(\rho_t, \mathbf{v_t})$ satisfies (CTY), then $(\rho_t, \mathbf{v_t} + \mathbf{w_t})$ also does for $\nabla \cdot (\mathbf{w}_t \rho_t) = 0$.

- $\|\mathbf{v} + \mathbf{w}\|_{L^2(\mu)} \geq \|\mathbf{v}\|_{L^2(\mu)} \; \forall \mathbf{w}$ s.t. $\nabla \cdot (\mathbf{w}\mu) = 0$ if and only if $\mathbf{v} \in \overline{\{\nabla \psi : \psi \in C_c^\infty(\mathbb{R}^d)\}}^{L^2(\mu)}$

Formal Riemannian structure

Thus, for $T_\rho M = \left\{ \partial_t \rho_t \Big|_{t=0} : \rho_t \text{ is a smooth curve in } M \text{ with } \rho_t|_{t=0} = \rho \right\}$,

the Benamou Brenier theorem suggests an inner product…

$$W_2^2(\rho_0, \rho_1) = \inf_{(\rho_t, \mathbf{v_t})} \left\{ \int_0^1 \int_\Omega |\mathbf{v_t}|^2 d\rho_t \, dt : \partial_t \rho_t + \nabla \cdot (\mathbf{v_t} \rho_t) = 0, \, \rho_t|_{t=0} = \rho_0, \, \rho_t|_{t=1} = \rho_1 \right\}$$

$$= \inf_{\rho_t} \left\{ \inf_{\mathbf{v_t} \,:\, \partial_t \rho_t = -\nabla \cdot (\mathbf{v_t} \rho_t)} \int_0^1 \int_\Omega |\mathbf{v_t}|^2 d\rho_t \, dt : \rho_t|_{t=0} = \rho_0, \, \rho_t|_{t=1} = \rho_1 \right\}$$

$$= \inf_{\rho_t} \left\{ \int_0^1 \int_\Omega |\nabla \psi_\mathbf{t}|^2 d\rho_t \, dt : \rho_t|_{t=0} = \rho_0, \, \rho_t|_{t=1} = \rho_1, \partial_t \rho_t = -\nabla \cdot (\nabla \psi_\mathbf{t} \rho_t) \right\}$$

$$= \inf_{\rho_t} \left\{ \int_0^1 \|\partial_t \rho\|_{\rho_t}^2 dt : \rho_t|_{t=0} = \rho_0, \, \rho_t|_{t=1} = \rho_1 \right\}$$

$$\left\langle h_1, h_2 \right\rangle_\rho = \int \nabla \psi_\mathbf{1} \cdot \nabla \psi_\mathbf{2} d\rho$$

$$h_i = -\nabla \cdot (\nabla \psi_i \rho)$$

Plan

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Gradient flows

$$\frac{d}{dt}x(t) = -\nabla_d E(x(t))$$

metric	definition of gradient	formula
$\mathbb{R}^d$	$\langle \nabla E(x), v \rangle = \lim_{h\to 0} \frac{E(x+hv)-E(x)}{h}$	$\nabla E(x) = \left[\frac{\partial E}{\partial x_i}\right]$
L^2	$\langle \nabla_{L^2} E(f), g \rangle_{L^2} = \lim_{h\to 0} \frac{E(f+hg)-E(f)}{h}$	$\nabla_{L^2} E(f) = \frac{\partial E}{\partial f}$
W_2	$\langle \nabla_{W_2} E(\rho_0), \partial_t \rho_t\vert_{t=0} \rangle_{\rho_0}$ $= \lim_{h\to 0} \frac{E(\rho_h)-E(\rho_0)}{h}$ $= [\ldots] = \left\langle -\nabla \cdot \left(\nabla \frac{\partial E}{\partial \rho_0} \rho_0 \right), \partial_t \rho_t\vert_{t=0} \right\rangle_{\rho_0}$	$\nabla_{W_2} E(\rho) = -\nabla \cdot \left(\nabla \frac{\partial E}{\partial \rho} \rho \right)$

W_2 gradient flows $\quad \partial_t \rho_t = -\nabla_{W_2} E(\rho_t)$

Fokker-Planck: [Jordan, Kinderlehrer, Otto '98,…]

$$\partial_t \rho = \nabla \cdot \left(\rho \nabla \log\left(\rho/\bar{\rho}\right) \right) = \Delta \rho - \nabla \cdot (\rho \nabla \log \bar{\rho}), \quad E(\rho) = \int \rho \log(\rho/\bar{\rho})$$

Aggregation + Drift: [McCann '97, Carrillo McCann Villani '05, Carrillo DiFrancesco, Figalli, Laurent, Slepcev '11,…]

$$\partial_t \rho = \nabla \cdot (\rho \nabla (K * \rho)) + \nabla \cdot (\rho \nabla V), \quad E(\rho) = \frac{1}{2} \iint K(x-y) d\rho(x) d\rho(y) + \int V \rho$$

2-layer neural networks: [Montanari Mei Nguyen '18, Rotskoff Vanden Eijnden '18, Chizat Bach '18, Sirigano Spiliopoulos '20,…].

$$E(\rho) = \frac{1}{2} \int \left| \int \Phi(x,z) d\rho(x) - f_0(z) \right|^2 d\nu(z)$$

Nonlinear diffusion equations:

$$\partial_t \rho = \nabla \cdot (\rho \nabla f'(\rho)) + \nabla \cdot (\rho \nabla V), \qquad E(\rho) = \int f(\rho) + \int V \rho$$

Particle solutions

For unif Lipschitz cts velocity $\mathbf{v} : \mathbb{R}^d \times \mathscr{P}_2(\mathbb{R}^d) \to \mathbb{R}^d$, consider

$$\begin{cases} \partial_t \rho_t(x) + \nabla \cdot (\rho_t(x)\ \mathbf{v}(x, \rho_t)) = 0, \\ \rho_t(x)\,|_{t=0} = \rho_0(x)\,. \end{cases}$$

1. Given initial data: $\rho_0^N := \dfrac{1}{N}\sum_{i=1}^{N} \delta_{x_i(0)}$
2. Evolve the locations:

$$\frac{d}{dt} x_i(t) = \mathbf{v}(x_i(t), \rho^N(t)) \iff \partial_t \rho^N + \nabla \cdot (\rho^N \mathbf{v}(\,\cdot\,, \rho^N)) = 0$$

$$\rho^N(t) = \frac{1}{N}\sum_{i=1}^{N} \delta_{x_i(t)}$$

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PDE reduces to system of ODEs

Particle method

Goal: Approximate solutions of PDE by **deterministic** evolution of empirical measures

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1. Approximate initial data: $\rho_0 \approx \rho_0^N := \dfrac{1}{N}\displaystyle\sum_{i=1}^N \delta_{x_i(0)}$

$$\rho^N(t) = \frac{1}{N}\sum_{i=1}^N \delta_{x_i(t)}$$

2. Evolve the locations:

$$\frac{d}{dt}x_i(t) = \mathbf{v}(x_i(t), \rho^N(t)) \iff \partial_t \rho^N + \nabla \cdot (\rho^N \mathbf{v}(\,\cdot\,, \rho^N)) = 0$$

3. Since $\mathbf{v}$ unif Lipschitz, $W_2(\rho^N(t), \rho(t)) \le e^{t\|\mathbf{v}\|_{\mathrm{Lip}}}\ W_2(\rho_0^N, \rho_0) \xrightarrow{N\to+\infty} 0.$

Nonlinear diffusion equations

$$\begin{cases} \partial_t \rho = \nabla \cdot (\rho \nabla f'(\rho)) + \nabla \cdot (\rho \nabla V), \\ \rho(x,0) = \rho_0(x) \end{cases}$$

Long-time behavior: If f is strictly convex, suff. regular, $s \mapsto s^d f(s^{-d})$ is convex and decreasing, and V with $D^2 V \geq \lambda I_{d\times d} > 0$,

$$W_2(\rho(t), \bar{\rho}) \leq e^{-\lambda t} W_2(\rho_0, \bar{\rho}), \qquad \bar{\rho} = \left((f^*)'(Z - V)\right)_+$$

Examples:

Fokker Planck	$f(a) = a\log(a) - a$	$f^*(b) = e^b$	$\bar{\rho} = e^Z e^{-V}$
Porous Medium	$f(a) = a^m/m$	$f^*(b) = b^{m'}/m'$	$\bar{\rho} = (Z - V)_+^{m'-1}$

$$\rho^N(t) := \frac{1}{N}\sum_{i=1}^N \delta_{X_i}(t) \xrightarrow{N\to+\infty} \rho(t) \xrightarrow{t\to+\infty} \bar{\rho}$$

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Langevin dynamics: $dX_t = \sqrt{2} dB_t + \nabla \log \bar{\rho}(X_t) dt$

$$\rho^N(t) := \frac{1}{N} \sum_{i=1}^N \delta_{X_i}(t) \xrightarrow{N\to+\infty} \rho(t) \xrightarrow{t\to+\infty} \bar{\rho}$$

Particle method for nonlinear diffusion

Nonlinear Diffusion Equation

$$\partial_t \rho = \nabla \cdot \left(\rho \nabla f'(\rho) \right) \qquad \mathscr{F}(\rho) = \int f(\rho)$$

Regularization

$$\partial_t \rho = \nabla \cdot \Big(\rho \underbrace{\nabla \varphi_\epsilon * \left(f'_\epsilon \left(\varphi_\epsilon * \rho \right) \right)}_{v_\epsilon[\rho]} \Big) \qquad \mathscr{F}_\epsilon(\rho) = \int f_\epsilon \left(\varphi_\epsilon * \rho \right)$$

Particle method for nonlinear

Nonlinear Diffusion Equation

$$\partial_t \rho = \nabla \cdot \left(\rho \nabla f'(\rho) \right) \qquad \mathscr{F}(\rho) = \int f(\rho)$$

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$\rho = \frac{1}{N} \sum_{i=1}^{N} \delta_{x_i}$

… implies…

$\varphi_\epsilon * \rho(x)$

$= \frac{1}{N} \sum_{i=1}^{N} \varphi_\epsilon(x - x_i)$

Particle method for nonlinear diffusion

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In Fokker Planck case, $v_\epsilon[\rho(t)] \approx \nabla \log \rho(t)$.

Particle method for nonlinear diffusion

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In Fokker Planck case, $v_\epsilon[\rho(t)] \approx \nabla \log \rho(t)$.

Simplified Theorem (C., Jacobs, Turanova, '25): Given particle initial data,

$$\rho_0^N := \frac{1}{N} \sum_{i=1}^{N} \delta_{x_i(0)} \xrightarrow{N \to +\infty} \rho_0,$$

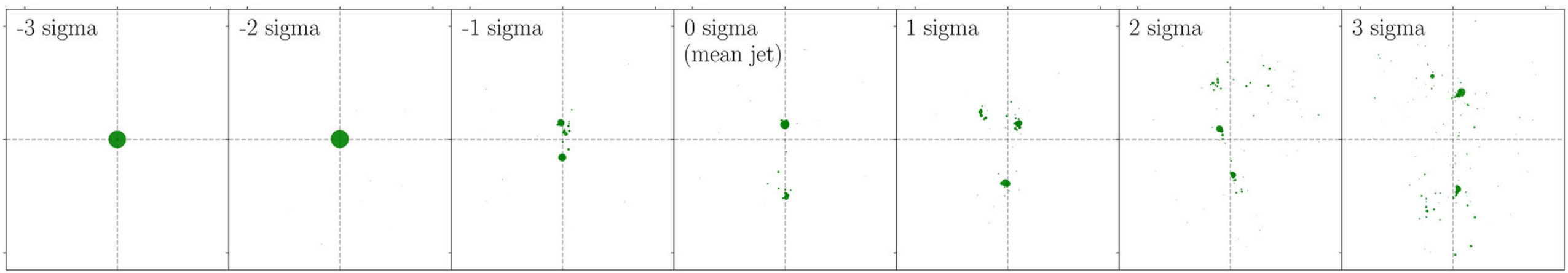
the particle solution $\rho_{N,\epsilon,t}$ satisfies

$$\rho_{N,\epsilon,t} \xrightarrow{N \to +\infty,\ \epsilon(N) \to 0} \rho_t \quad \forall t \geq 0.$$

See also [Carrillo, Esposito, Skrzeczkowski, Wu '24] for m = 1 and m<1.

Plan

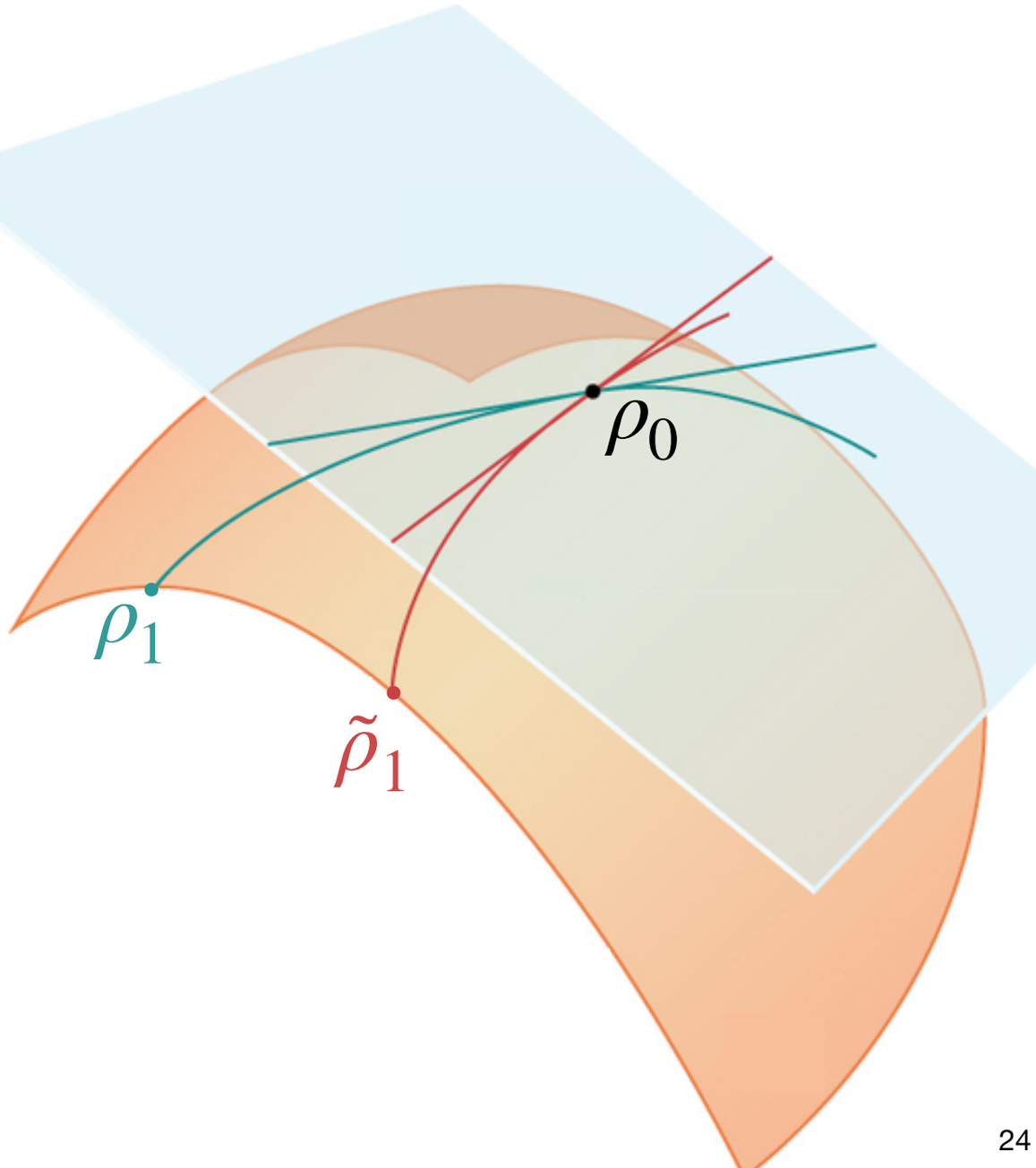
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Linearized 2-Wasserstein Metric

[Wang, Slepčev, Basu, Ozolek, Rohde '13]

Recall: If $\rho_t = \mathbf{T}_t \# \rho_0$, then $\mathbf{v_t} \circ \mathbf{T}_t = \mathbf{T}_{\rho_0}^{\rho_1} - \mathbf{I} \Longrightarrow \mathbf{v_0} = \mathbf{T}_{\rho_0}^{\rho_1} - \mathbf{I} = \nabla \psi$

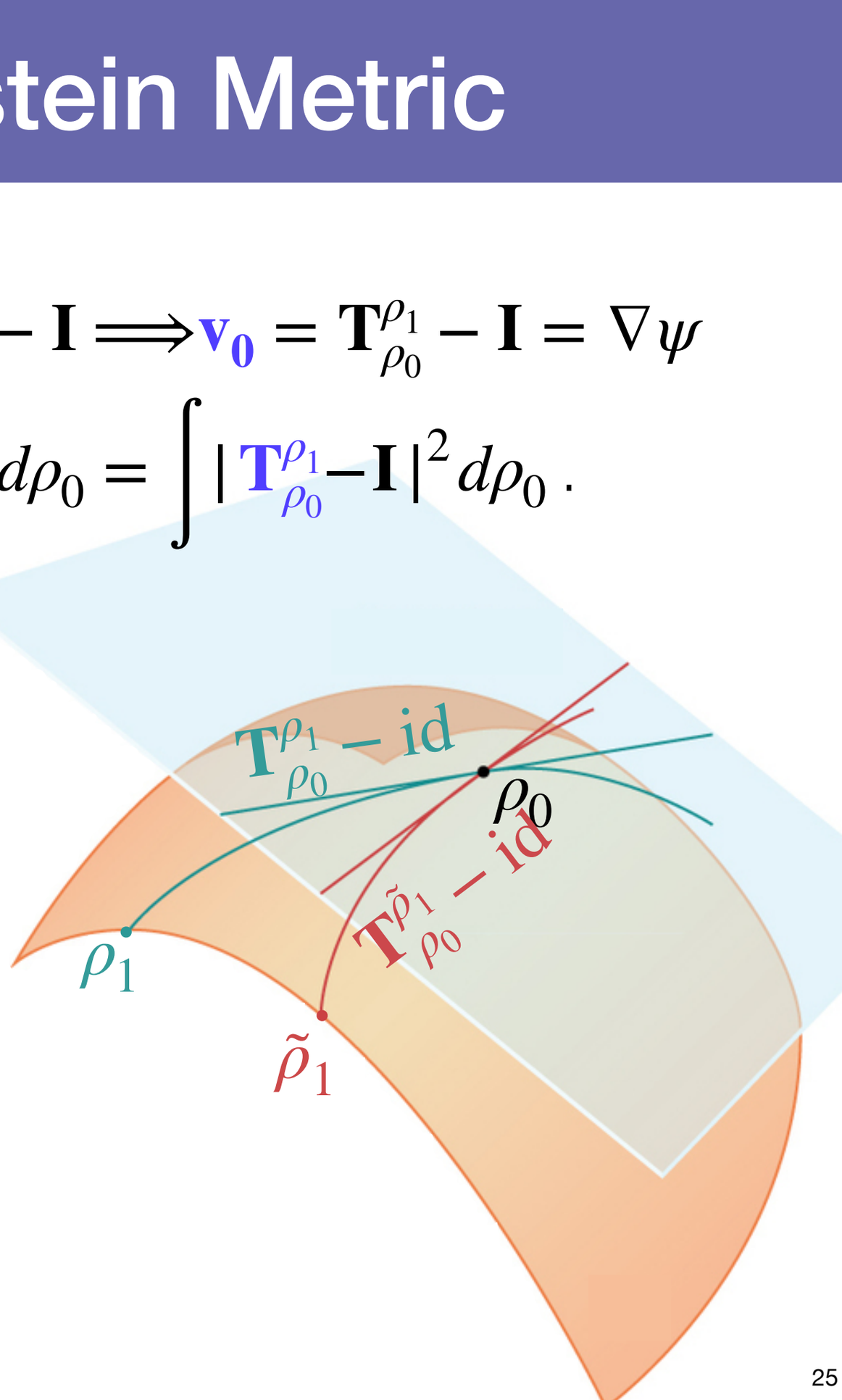


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In this case, $\left\| \partial_t \rho_t |_{t=0} \right\|_{\rho_0}^2 = \int |\nabla \psi|^2 d\rho_0 = \int |\mathbf{T}_{\rho_0}^{\rho_1} - \mathbf{I}|^2 d\rho_0 .$



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[Wang, Slepčev, Basu, Ozolek, Rohde '13]

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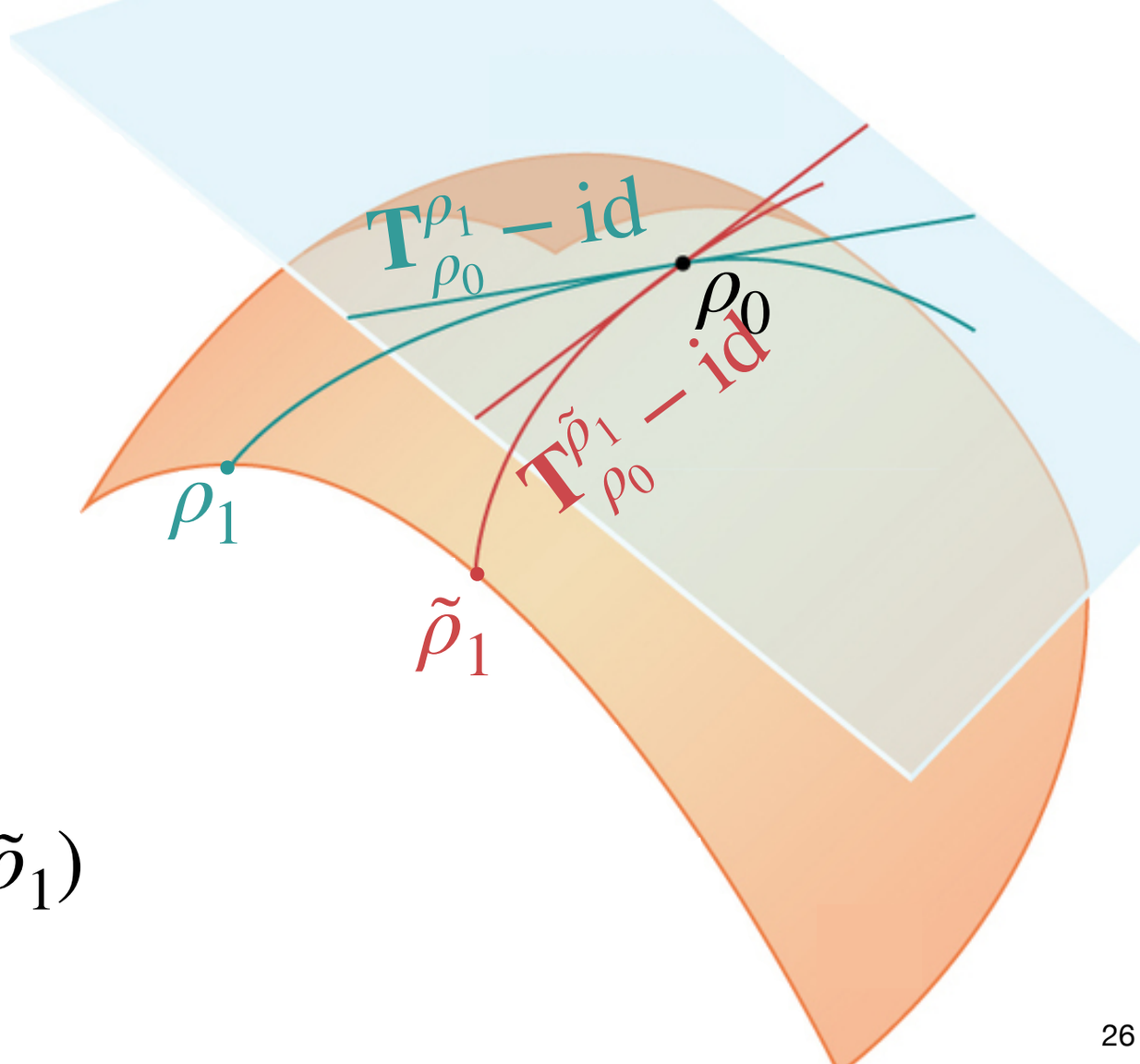
Thus, to linearize W_2 about ρ_0:

$$W_2^2(\rho_1, \tilde{\rho}_1)$$

$$\approx \left\| \partial_t \rho_t |_{t=0} - \partial_t \tilde{\rho}_t |_{t=0} \right\|_{\rho_0}^2$$

$$= \int |(\mathbf{T}_{\rho_0}^{\rho_1} - \mathbf{I}) - (\mathbf{T}_{\rho_0}^{\tilde{\rho}_1} - \mathbf{I})|^2 d\rho_0$$

$$= \int |\mathbf{T}_{\rho_0}^{\rho_1} - \mathbf{T}_{\rho_0}^{\tilde{\rho}_1}|^2 d\rho_0 =: \mathrm{LOT}_{\rho_0}(\rho_1, \tilde{\rho}_1)$$



Linearized 2-Wasserstein Metric

[Wang, Slepčev, Basu, Ozolek, Rohde 2013]

$$\mathrm{LOT}_{\rho_0}(\rho_1, \tilde{\rho}_1) = \int |\mathbf{T}_{\rho_0}^{\rho_1} - \mathbf{T}_{\rho_0}^{\tilde{\rho}_1}|^2 d\rho_0$$

Benefits of LOT

- LOT_{ρ_0} is a metric. Also,

  $W_2(\mu,\nu) \le \mathrm{LOT}_{\rho_0}(\mu,\nu) \le C W_2(\mu,\nu)^{1/6}$

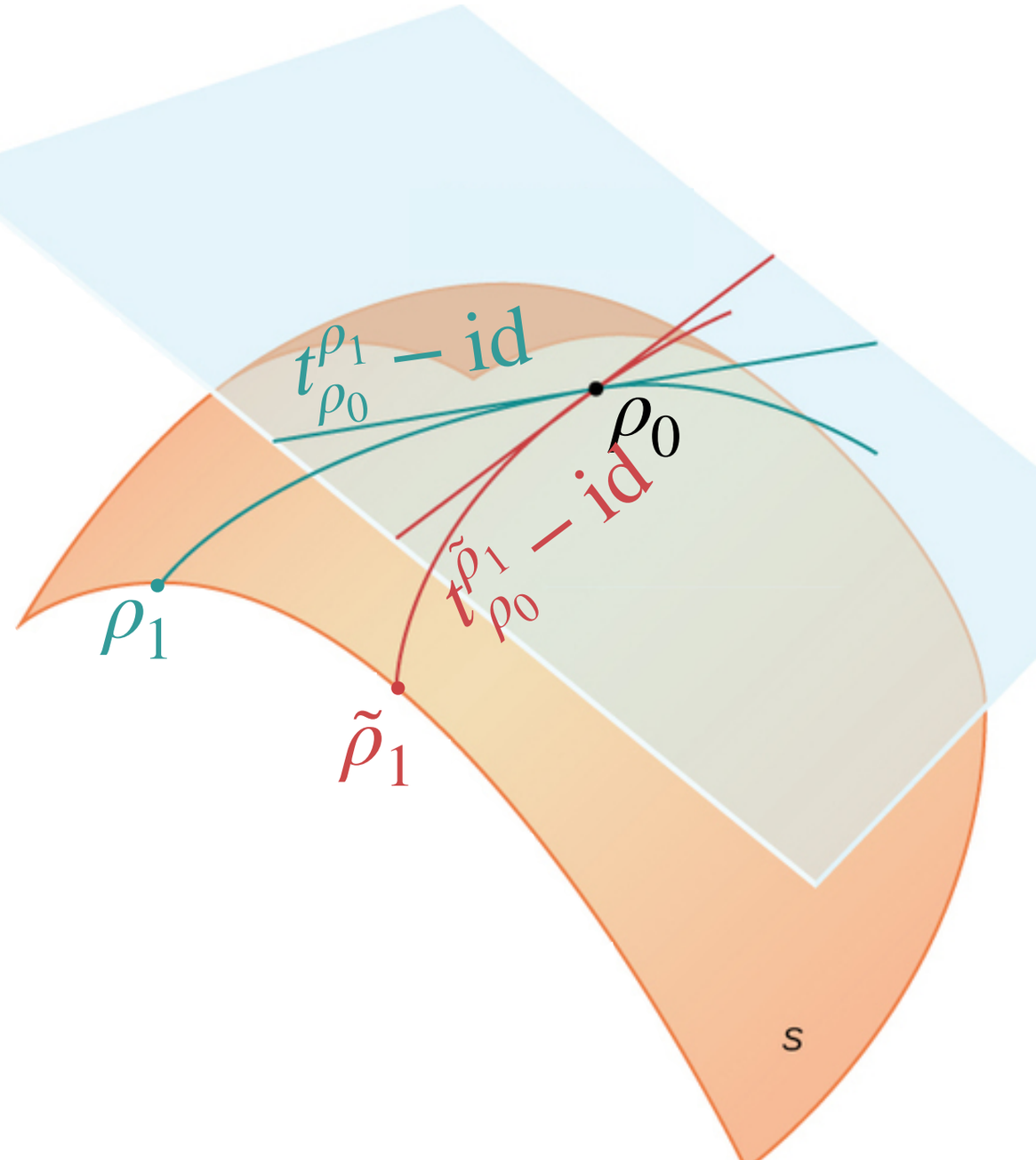
  [Delalande, Mérigot 2020, 2021]

  [Letrouit, Mérigot '24],…

- Computing $\mathbf{LOT}_{\sigma}$ between $\mathcal{O}(N)$ images requires $\mathcal{O}(N)$ OT calcs and $\mathcal{O}(N^2)$ $L^2(\rho_0)$ calcs.

- Embedding into linear space

  $\mu \mapsto \mathbf{T}_{\rho_0}^{\mu} - \mathrm{id}$



Linearized 2-Wasserstein Metric

[Wang, Slepčev, Basu, Ozolek, Rohde 2013]

$$\mathrm{LOT}_{\rho_0}(\rho_1, \tilde{\rho}_1) = \int |\mathbf{T}_{\rho_0}^{\rho_1} - \mathbf{T}_{\rho_0}^{\tilde{\rho}_1}|^2 d\rho_0$$

Benefits of LOT

- LOT_{ρ_0} is a metric. Also,

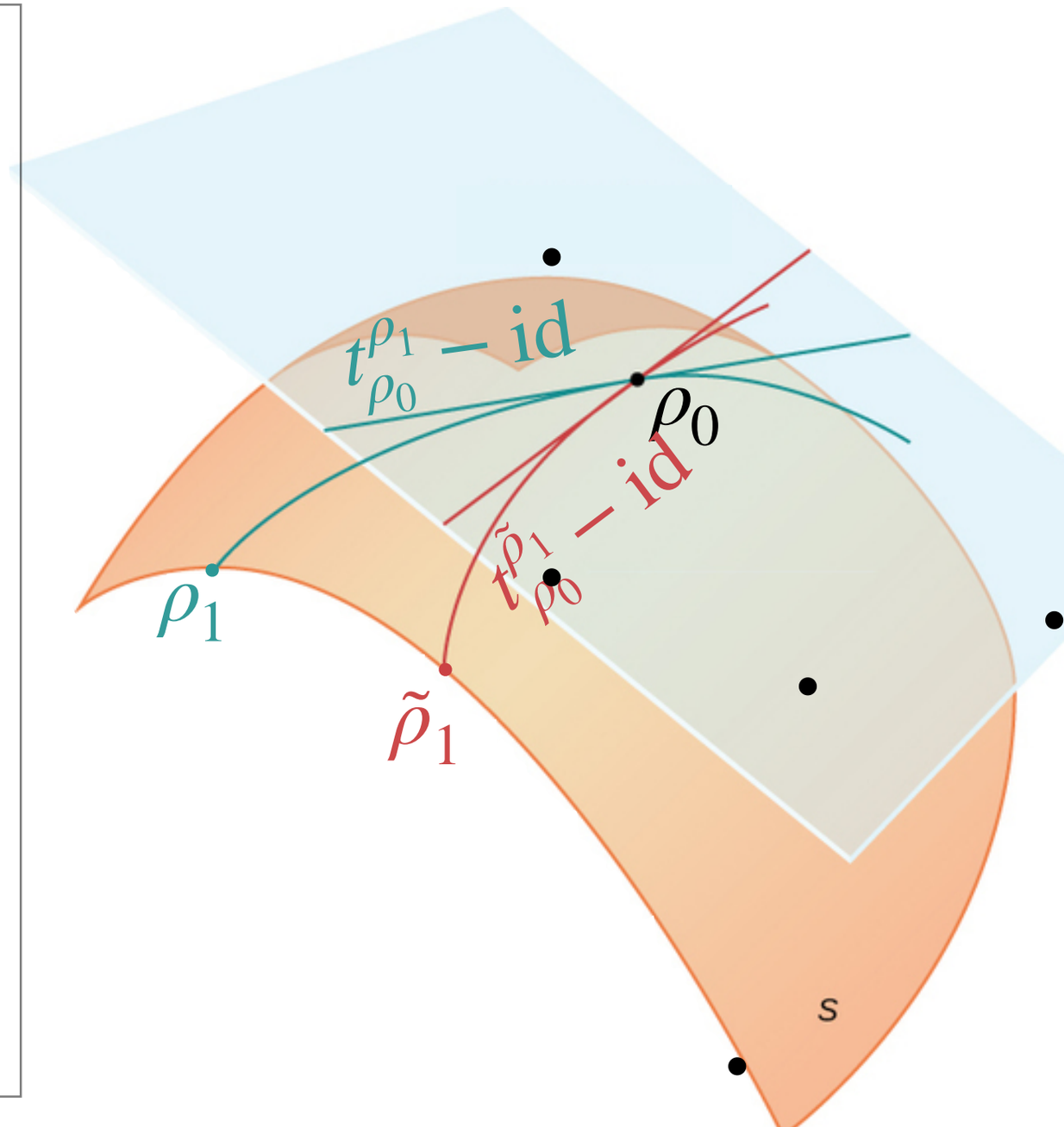
  $W_2(\mu,\nu) \le \mathrm{LOT}_{\rho_0}(\mu,\nu) \le C W_2(\mu,\nu)^{1/6}$

  [Delalande, Mérigot 2020, 2021]

  [Letrouit, Mérigot '24],…

- Computing $\mathbf{LOT}_\sigma$ between $\mathcal{O}(N)$ images requires $\mathcal{O}(N)$ OT calcs and $\mathcal{O}(N^2)$ $L^2(\rho_0)$ calcs.
- Embedding into linear space

  $\mu \mapsto \mathbf{T}_{\rho_0}^{\mu} - \mathrm{id}$



Linearized 2-Wasserstein Metric

[Wang, Slepčev, Basu, Ozolek, Rohde 2013]

$$\mathrm{LOT}_{\rho_0}(\rho_1, \tilde{\rho}_1) = \int |\mathbf{T}_{\rho_0}^{\rho_1} - \mathbf{T}_{\rho_0}^{\tilde{\rho}_1}|^2 d\rho_0$$

Benefits of LOT

- LOT_{ρ_0} is a metric. Also,

  $W_2(\mu,\nu) \le \mathrm{LOT}_{\rho_0}(\mu,\nu) \le C W_2(\mu,\nu)^{1/6}$

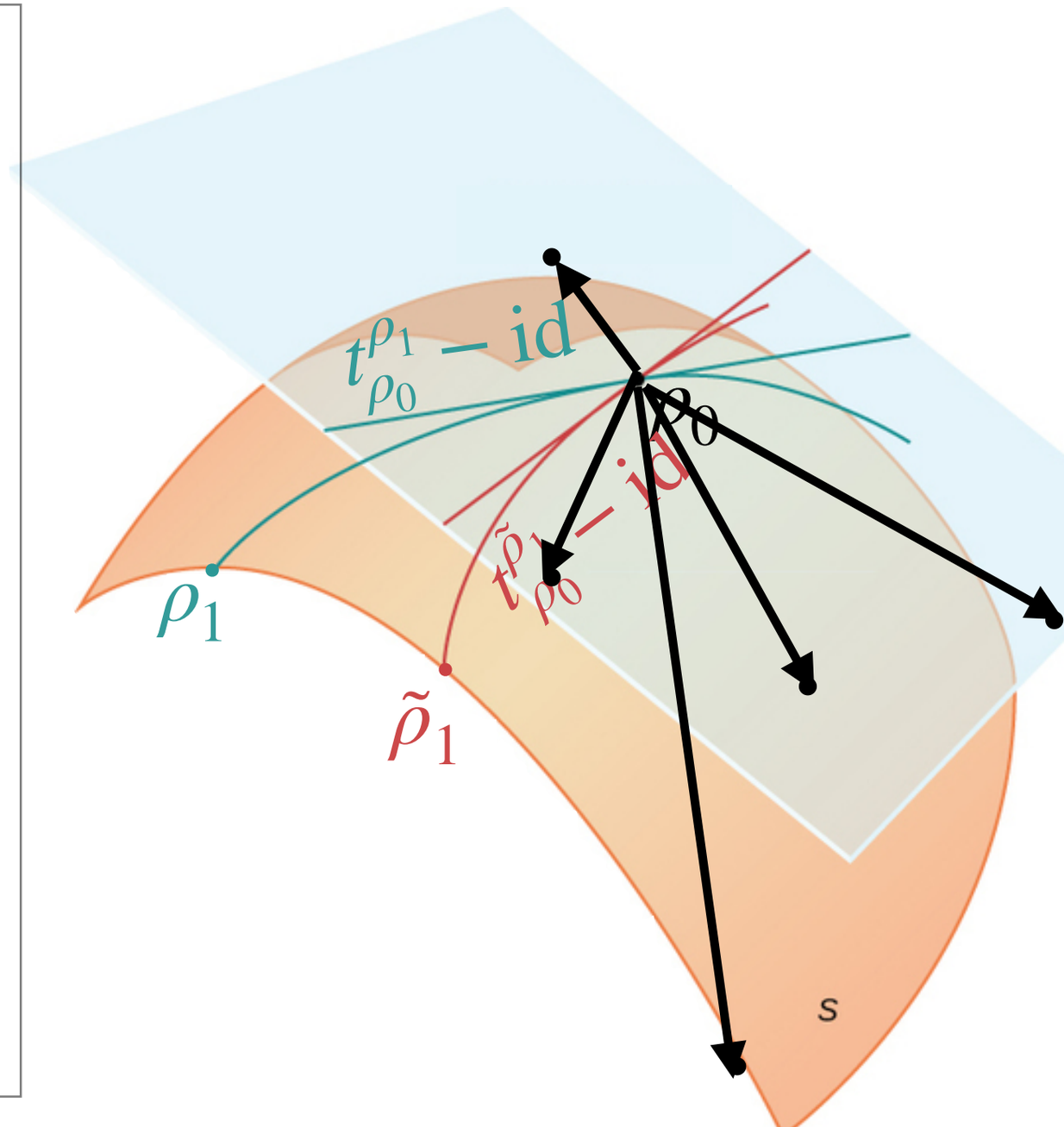
  [Delalande, Mérigot 2020, 2021]

  [Letrouit, Mérigot '24],…

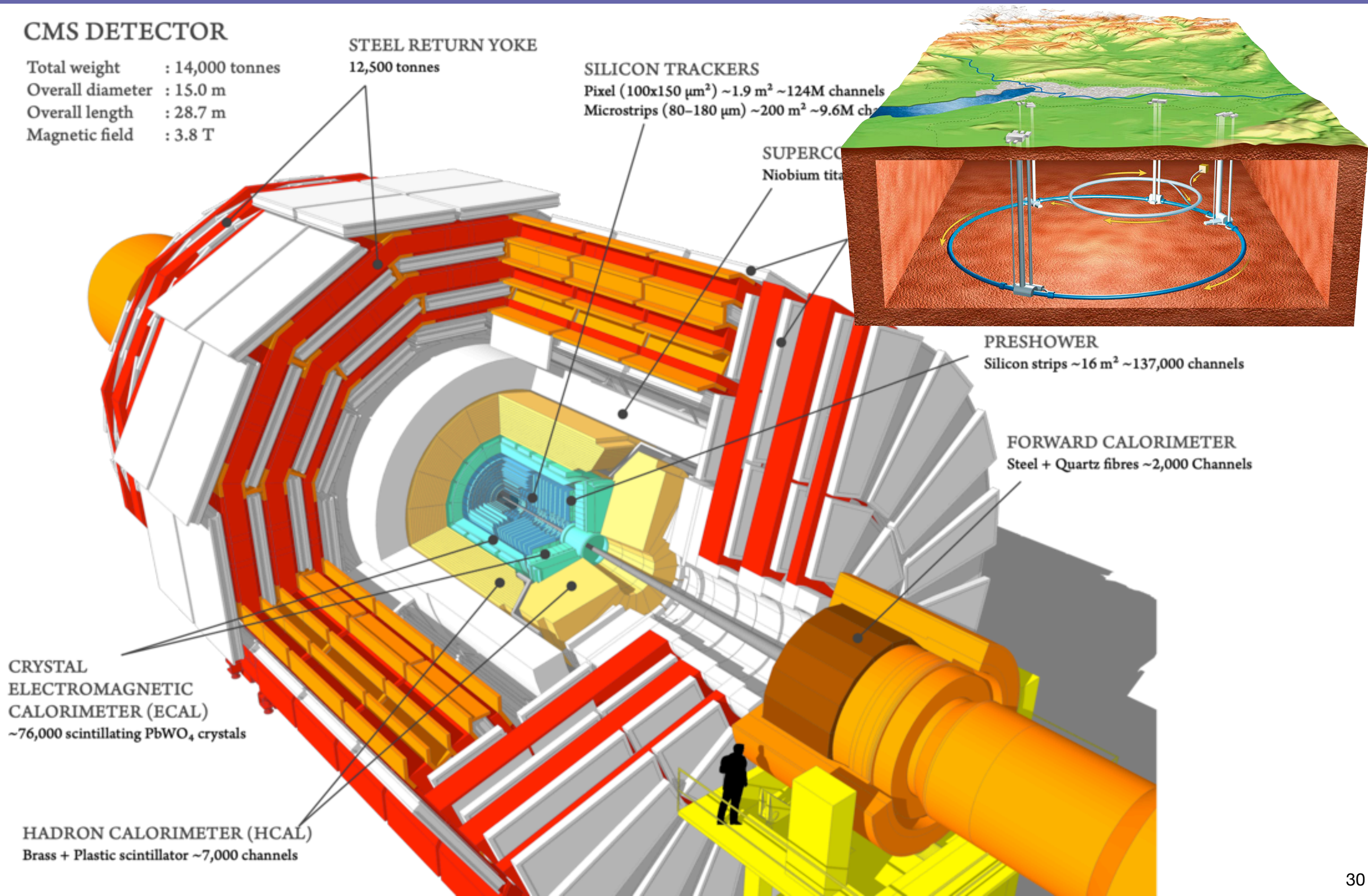
- Computing $\mathbf{LOT}_\sigma$ between $\mathcal{O}(N)$ images requires $\mathcal{O}(N)$ OT calcs and $\mathcal{O}(N^2)$ $L^2(\rho_0)$ calcs.

- Embedding into linear space

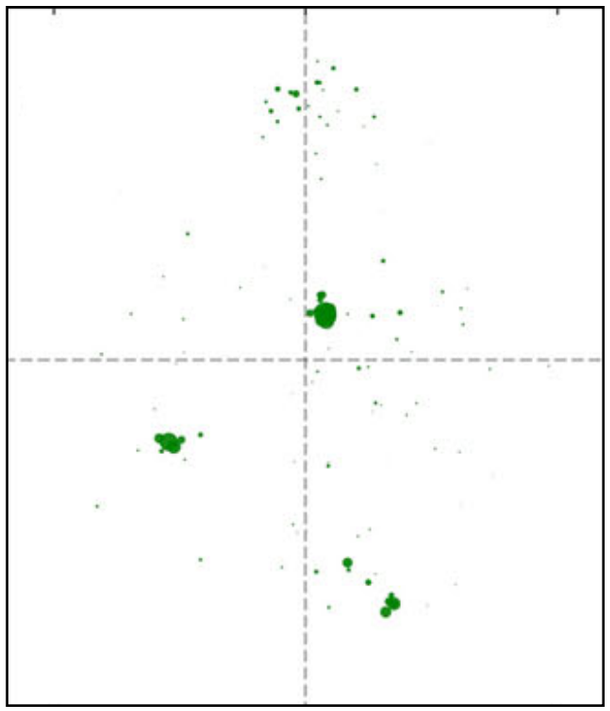
  $\mu \mapsto \mathbf{T}_{\rho_0}^{\mu} - \mathrm{id}$



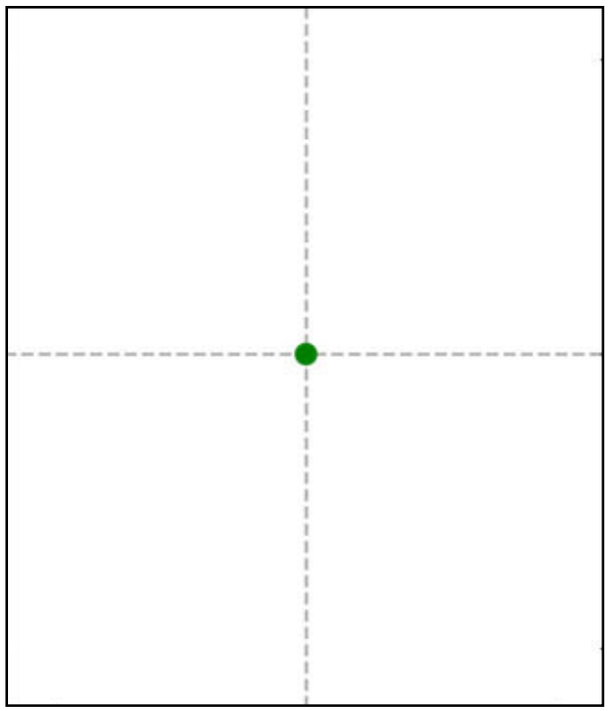
Application: Large Hadron Collider



Jet tagging



t^1 jet



W^1 jet

Key features of hCal data:

- spatial location is meaningful
- minimal overlapping support

normalization…

Goal: use hCal measurements to infer what occurred at the parton level.

Previous Work [Komiske, Metodiev, Thaler, 2019] used W_1 metric

- **Pros:** OT outperforms classical methods, approaches NN accuracy with superior interpretability
- **Cons:** ~16 years on a laptop; large storage burden; limited classification algorithms.

Event classification with LOT

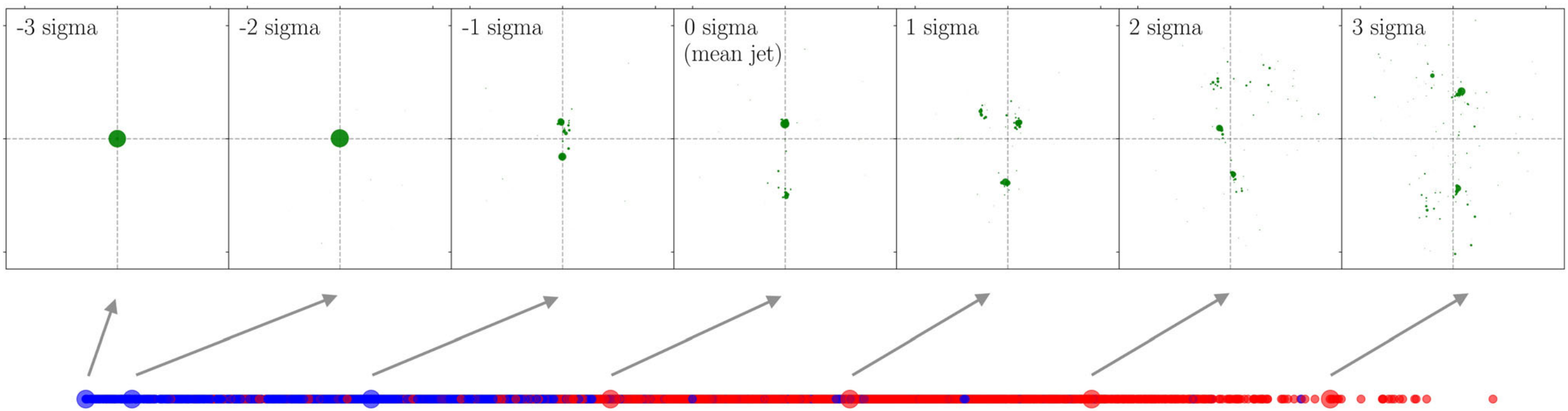
[Cai, Cheng, Craig, Craig 2020]

- W vs QCD jets
  - balanced 100K event training sample, 20K event test sample

- Computation of LOT embedding requires ~ 20 minutes on a laptop.

- Interpretable:

Datasets	**Model**	**AUC**
Our work	$\mathrm{k}_{=20}$NN-LOT	0.845
	SVM-LOT	0.869
	LDA-LOT	0.704
[Komiske, Metodiev, Thaler, 2019]	$\mathrm{k}_{=32}$NN-EMD	0.887
	$\tau_2^{\beta=1}$ / $\tau_1^{\beta=1}$	0.776
	PFN	0.919
	EFPs	0.917
	EFN	0.904



Conclusion

Gradient Flows and Sampling

- With formal Riemannian structure, one can define 2-Wasserstein gradient flows.
- Classical results used GF structure to study approximation/ long time behavior of diffusion equations…
- … more recently, can use GF structure to define novel deterministic algorithms for sampling non-log concave measures.

ML

- With formal Riemannian structure, one can linearize 2-Wassersein metric about a reference measure
- Applications in particle physics demonstrate this leads to a huge efficiency gains, with only marginal degradation in accuracy…
- … many open questions about the design of reference measure and the interplay between LOT and classification algorithms.

Thank you!